YoungJo Choi (Jo Choi)

Jo Choi | Brooklyn, New York, USA | yc7330@nyu.edu | +1-312-918-9643 | **Github**: TechJoe96
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EDUCATION

New York University Fall 2024-Present
- M.S., Computer Engineering (Expected graduation date: May 2026)

Illinois Institute of Technology (Dual Degree program) Fall 2022-Spring 2024
- B.S., Computer Engineering, minor in Artificial Intelligence

Inha University (Dual Degree program) Spring 2016-Spring 2024
- B.S., Electronic Engineering

SKILLS

- **Programming&Scripting:** SystemVerilog, Verilog, VHDL, RISC-V Assembly, ARM Assembly, MATLAB, Python, C++, C, Java, PostgreSQL, MySQL, R
- **Design Tools:** Cadence Virtuoso/Genus/Innovus, Xilinx Vivado, ModelSim, KiCAD, AutoCAD
- **AI/ML Frameworks:** PyTorch, TensorFlow, Scikit-learn, Keras, MLflow
- **Data & Visualization:** Pandas, NumPy, PowerBI, Tableau, Excel, Selenium
- **Other:** React, HTML/CSS/JavaScript, Node.js, REST API, FastAPI, Flask, SageMaker AWS, Docker, Kubernetes, Linux, Git

WORK EXPERIENCE

Teaching Assistant at New York University Sep 2025-Dec 2025
- Teaching lab sections for ECE-UY 3114 Fundamentals of Electronics I at New York University
- Ran experiments across amplifiers, diodes and transistor circuits

Data Analyst Intern at **C&B Trading** in Seoul, Korea Jun 2024-Aug 2024
- Improved sales forecast accuracy to 90% with a predictive model; built Excel reports and PowerBI dashboards that identified three key process bottlenecks for senior management.

Republic of Korea (ROK) Military Aug 2017-Apr 2019
- Honorably discharged as a Sergeant and a Squad Leader after serving in ROK for 21 months

RESEARCH EXPERIENCE & PROJECTS

OpenPOWER SoC Design with Microwatt CPU and NN Accelerator Fall 2025
- FPGA/ASIC-ready neural network accelerator (Verilog RTL) featuring INT8 systolic array processing for OpenPOWER systems

GenAI-based Chip Design Fall 2025

1) Developing LLM model architectures for automating chip design across RTL, verification to physical design, also for design optimization under the guidance of Prof. Ramesh Karri
2) Designing a Quantized Neural Net Accelerator with the Microwatt OpenPOWER CPU on SKY130, enabling efficient ML inference via memory-mapped I/O and custom instruction paths

Efficient Vision-Language Model (VLM) Design with Sparse Techniques Sep 2025-Present

1) Developing a saliency-based VLM by integrating a Vision Transformer with an LLM through a projection layer, enabling efficient visual token selection and multimodal reasoning under the guidance of Prof. Sai Qian Zhang
2) Reproduced SparseVLM on LLaVA-1.5-7B, reducing visual tokens by 30% with minimal accuracy loss; outperformed three weight-pruning baselines on VQAv2

Superconducting Nanoribbon Device Fabrication Mar 2025-Aug 2025
- Developed superconducting nanoribbon wires on HbN substrate at Nano Lab under the guidance of Prof. Davood Shahrjerdi
- Engineered the recipes of E-beam lithography, Deep Reactive-Ion Etching, and Buffered Oxide Etching processes in cleanroom environment

Real-Time Parkinson's Tremor & Dyskinesia Detector Spring 2025
- Built a real-time embedded system on STM32L475 that detects Parkinson's tremor (3–5 Hz) and dyskinesia (5–7 Hz) using accelerometer data and FFT-based frequency analysis
- Implemented a signal processing pipeline (DC offset removal, Hann windowing, 256-point FFT) capturing 156 samples at 52 Hz in 3-second analysis windows with real-time LED feedback

Mouse Motion Tracker using Deep Learning at IIT Sep 2023-May 2024
- Developed an RNN-based multimodal system using visual and audio data to recognize six mouse behaviors with 90% accuracy under the guidance of Prof. Yan Yan

LEADERSHIP EXPERIENCE

- Active member at **Triangle fraternity (Armour Chapter)** Fall 2022-Spring 2024
- **President & Founder** of Korean Student Association at Illinois Institute of Technology Summer 2023-Spring 2024